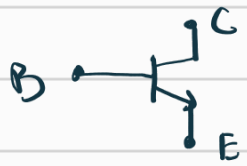
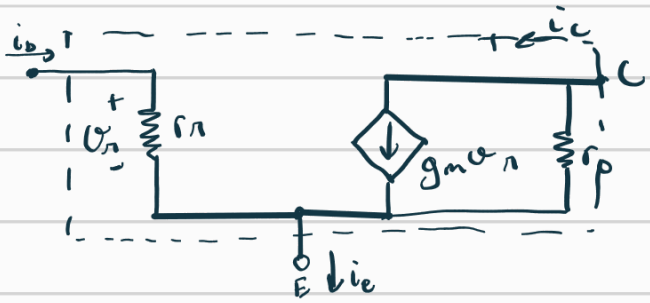
# Small signal Models




## π model

$v_c \ll V_T \rightarrow 0.025V$




$$i_b = \frac{v_\pi}{r_\pi}$$

**Small signal model:** $r_o \rightarrow$ output resistance of bjt

$$r_o = \frac{|V_A|}{I_C}$$

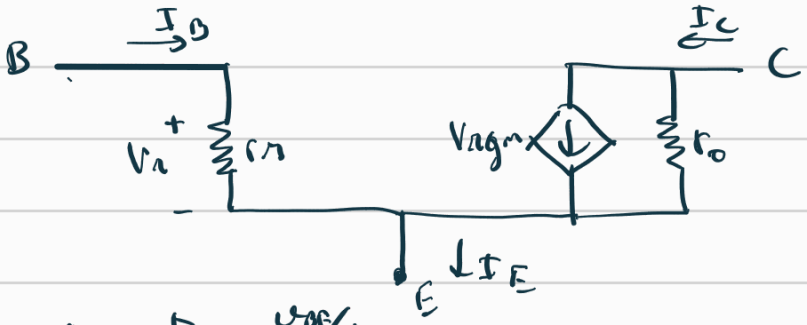
$g_m \rightarrow$ transcunductance $= g_m = \frac{I_C}{V_T}$

$$\alpha = \frac{\beta}{(\beta+1)}$$

$r_\pi \rightarrow$ base resistance $= r_\pi = (\beta+1)\, r_e$

$r_e \rightarrow$ emitter resistance $= \frac{V_T}{I_E} = \alpha \frac{V_T}{I_C} = \alpha \frac{1}{g_m} \approx \frac{1}{g_m}$

$\alpha \approx 1$ //




| | |
|---|---|
| $g_m = \frac{I_C}{V_T}$ | $r_\pi = (\beta+1) r_e$ // |
| $r_o = \frac{\lvert V_A \rvert}{I_C}$ | $r_e = \frac{V_T}{I_E} = \frac{\alpha V_T}{I_C} = \frac{\alpha}{g_m}$ |

$$i_C = I_S e^{v_{BE}/V_T}$$

$$i_B = \frac{i_C}{\beta}$$

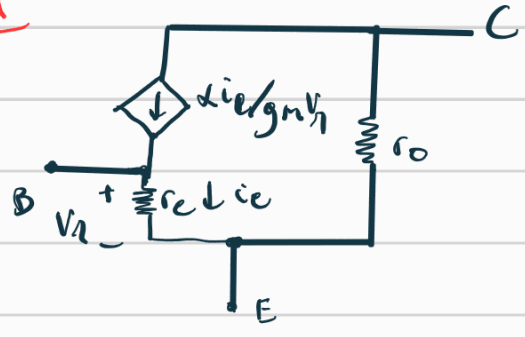
$$i_E = \frac{\beta+1}{\beta} i_C = (\beta+1) i_B$$

$$r_\pi i_B = r_e i_E$$

$$r_\pi i_B = r_e (\beta+1) i_B$$

$$r_\pi = (\beta+1) r_e$$

## T model

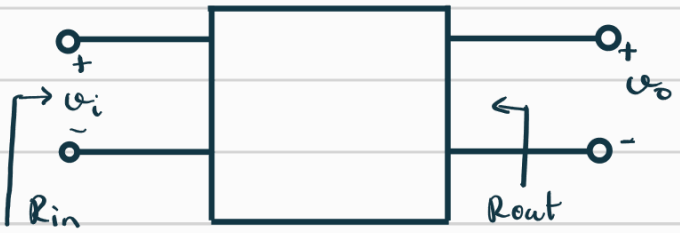



$$r_o = \frac{|V_A|}{I_C} \qquad \alpha = \frac{\beta}{\beta+1}$$

$$r_e = \frac{\alpha}{g_m} \qquad \alpha = r_e g_m$$

$$V_\pi = r_e \cdot i_e \qquad i_e = \frac{V_\pi}{r_e}$$

$$\alpha i_c = \alpha g_m \cdot \frac{V_\pi}{r_e}$$




## Voltage Gain

$$A_v = \frac{v_o}{v_i}$$

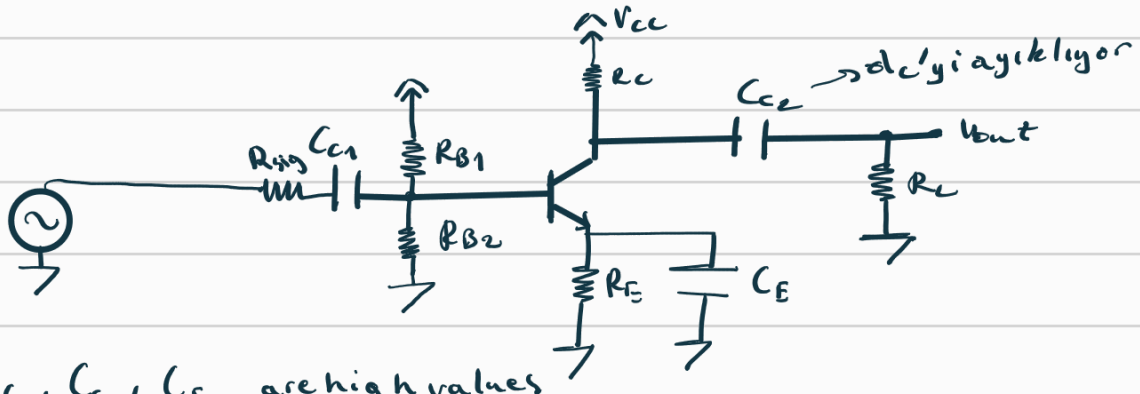
$R_{in}$ → Input resistance

$R_{out}$ → Output

## Amplifier Topologies

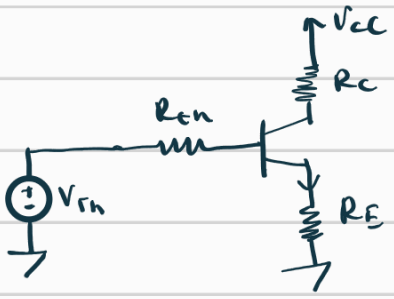
- ✱ Common Emitter
- ✱ Common Emitter with $R_E$
- ✱ Common Base
- ✱ Common-Collector (Emitter-Follower)

## Common Emitter Amplifier




$C_{c1}$, $C_{c2}$, $C_E$ are high values

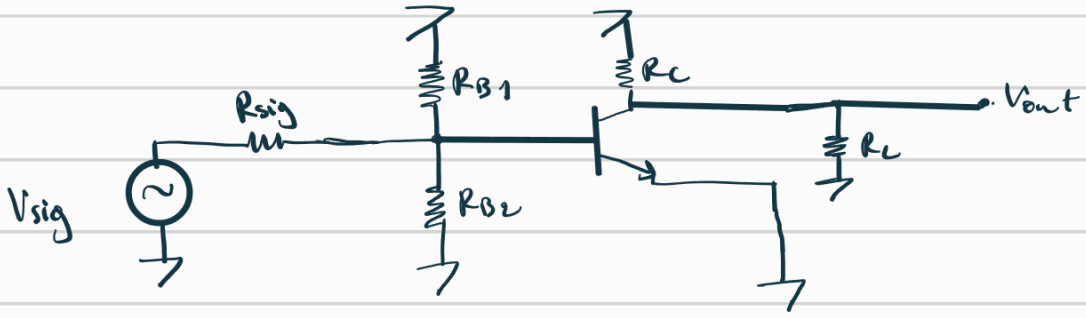
C are open circuit at DC ac sources short circuit



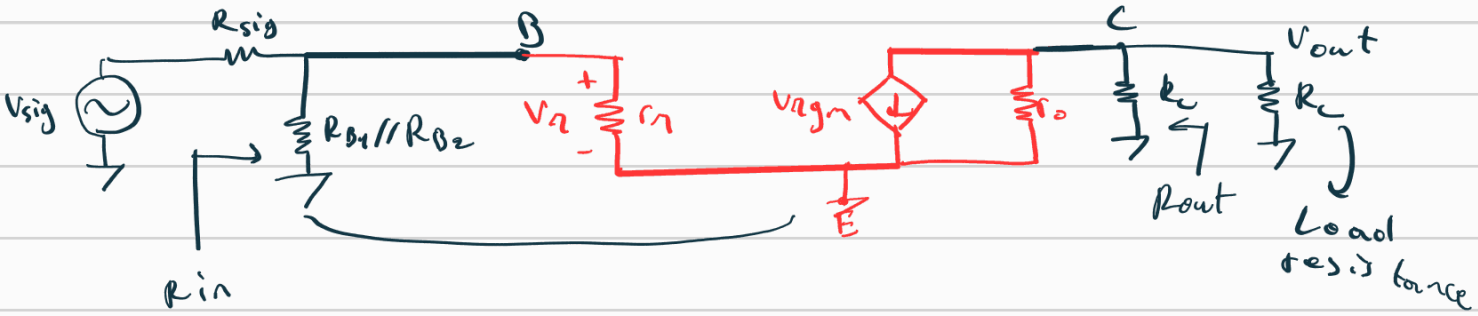

$g_m, r_e, r_\pi, r_o$

@ AC

- Now all caps are short circuit
- DC voltage sources are short circuit
- DC current sources are open circuit



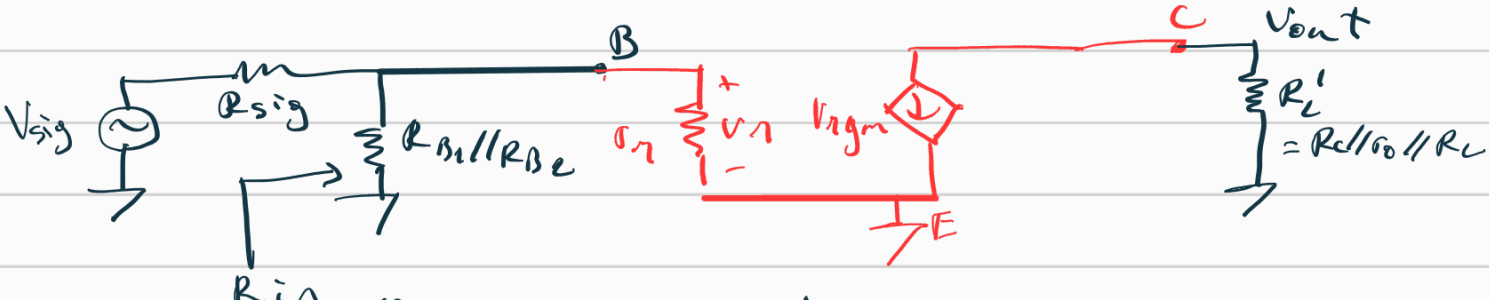

Small signal circuit




$$A_v = \frac{v_{out}}{v_{in}}, R_{in}, R_{out}$$

$$R_{in} = R_{B1} // R_{B2} // r_\pi$$

$$R_{out} = R_C // r_o$$




$$v_{out} = -g_m V_\pi \cdot R_L'$$

$$v_{in} = v_{sig}$$

$$v_\pi = \frac{v_{sig} \cdot (R_{B1} // R_{B2} // r_\pi)}{R_{sig} + (R_{B1} // R_{B2} // r_\pi)}$$

$$\frac{v_{out}}{v_{sig}} = \frac{-g_m v_{\pi} \cdot R_L'}{v_{sig}}$$

$$= -g_m \left( \frac{v_{sig} R_{B1} // R_{B2} // r_\pi}{v_{sig}\ R_{sig} + (R_{B1} // R_{B2} // r_\pi)} \right) R_L'$$

$R_{in}$

if $R_{sig} \approx 0 \Rightarrow = -g_m R_L'$

→ High Voltage Gain $\approx -g_m R_L'$

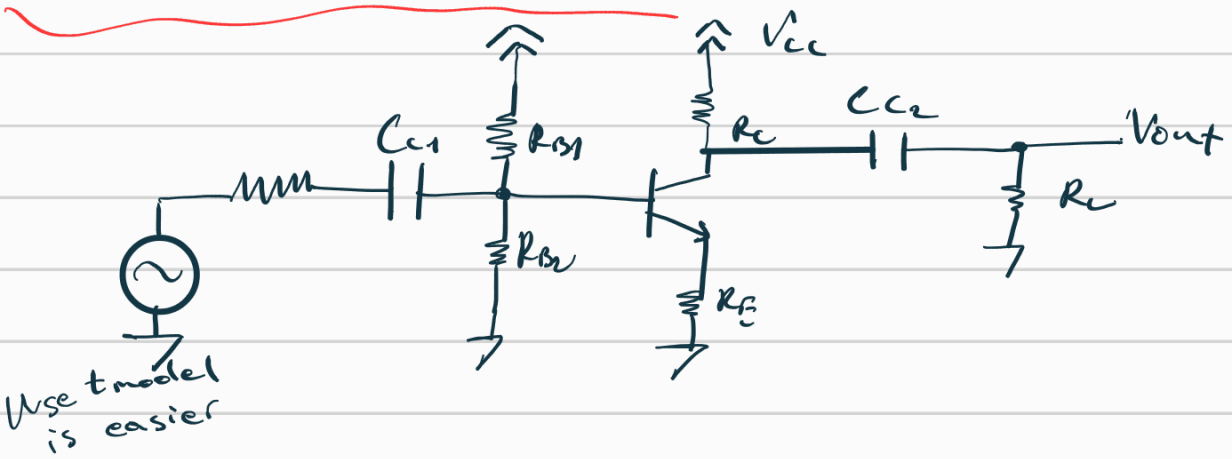
→ Moderate Input Resistance → $R_{B1} // R_{B2} // r_\pi$

↳ Moderate High output Resistance → $R_C // r_o$

Rin'in büyük olmasını isteriz çok büyük olursa Rsig ihmaledilebilecek kadar küçük olur. ve daha yüksek kazanç sağlanır.

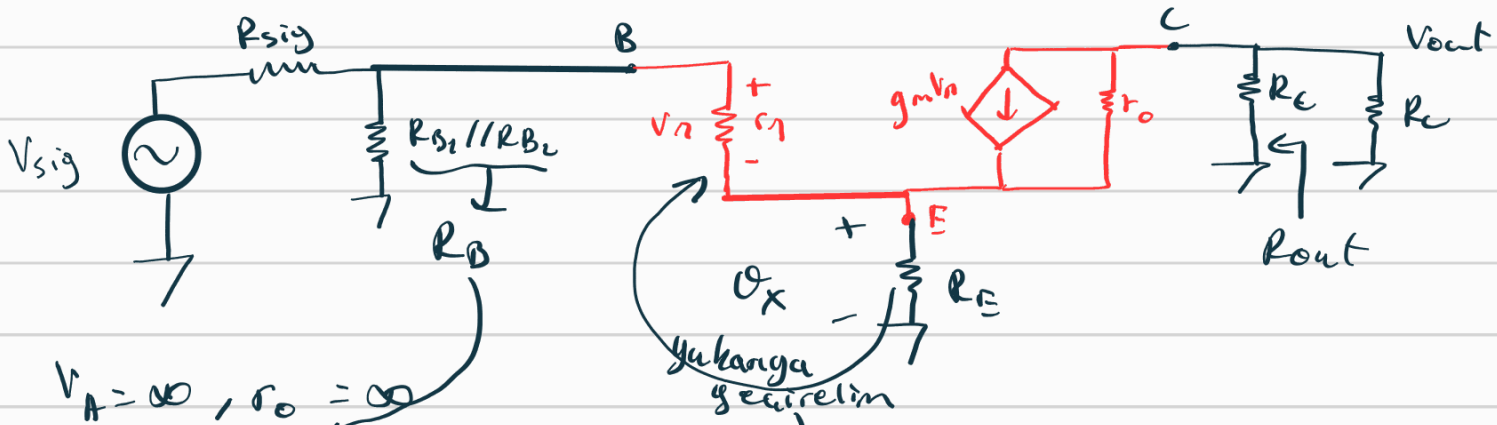
Routun küçük olmasını isteriz böylece üzerinde voltaj düşümü azalır ve sinyal daha kazançlı bir şekilde çıkışa aktarılabilir.

## Common Emitter Amplifier with $R_E$




Use t model is easier

## with r model




$V_A = \infty$ , $r_o = \infty$

$$R_{in} = R_B // \left( r_\pi + R_E (\beta + 1) \right)$$

$$= R_B // \left[ (\beta + 1)(r_e + R_E) \right]$$

$$R_{out} \cong R_C$$

$$A_v = \frac{v_{out}}{v_{in}}$$

$$v_{out} = -g_m V_\pi (R_C // R_L) \qquad V_\pi \stackrel{?}{=} V_{sig}$$

$$v_x = \left( g_m V_\pi + \frac{V_\pi}{r_\pi} \right) R_E$$

$$= \left( \frac{V_\pi}{r_e} + \frac{V_\pi}{r_\pi} \right) R_E \cong \frac{v_\pi}{r_e} \cdot R_E$$

$$r_e = \frac{V_T}{I_E} = \frac{\alpha V_T}{I_C} = \frac{\alpha}{g_m}$$

$$v_b = v_\pi \left( 1 + \frac{R_e}{r_e} \right)$$

if $R_{sig} = 0$ $\quad v_b = v_{sig}$

$$\frac{v_{out}}{v_{in}} = \frac{-g_m V_\pi (R_C // R_L)}{V_\pi \left( 1 + \frac{R_E}{r_e} \right)}$$

if $R_{sig} = 0$ $\quad v_b = v_{sig}$

$$\frac{v_{out}}{v_{in}} = \frac{-g_m V_\pi (R_C // R_L)}{V_\pi \left( 1 + \frac{R_E}{r_e} \right)} \qquad r_e = \frac{1}{g_m}$$

$$\frac{v_{out}}{v_{in}} = \frac{-g_m (R_C \parallel R_L)}{r_e \left(1 + \frac{R_E}{r_e}\right)}$$

$$\approx -\frac{(R_C \parallel R_L)}{r_e + R_e}$$

$$A_v \approx -\frac{R_C \parallel R_L}{R_E + r_e}$$

